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RESEARCH NOTE

UNDERSTANDING THE RETURNS FROM INTEGRATED ENTERPRISE SYSTEMS: THE IMPACTS OF AGILE AND PHASED IMPLEMENTATION STRATEGIES[1]

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How do firms benefit from integrated enterprise systems (IES), and how does the IES implementation strategy influence the returns from IES? We investigated the implementation strategy that firms should follow to integrate multiple enterprise systems regarding the timing of adoption and the richness of modules. Borrowing theories from software development literature and enterprise system implementation literature, we developed two IES implementation strategies, i.e., the agile strategy (simple, quick, and flexible) and the phased strategy (rich, phased, and pre-determined). We collected a sample of 675 public firms from 1997 to 2004 at the module level, enabling us to distinguish enterprise resource planning (ERP) systems from customer relationship management (CRM) and supply chain management (SCM) systems. Our data contained the timing of purchase and "go-live" events of these system modules, helping us understand firms' detailed implementation decisions. We found that IES returns depend on the choices of ERP and CRM/SCM module foundation, ERP-CRM/SCM sequential connection, and the continued adoption of ERP and CRM/SCM modules. Fewer ERP or CRM/SCM modules at CRM/SCM go-live events, a quicker connection between ERP and CRM/SCM go-live events, and the continued adoption of ERP or CRM/SCM modules all were found to enhance IES returns. Our findings show that the agile strategy leads to more returns from IES than the phased strategy and suggest that firms should integrate multiple enterprise systems by going live with advanced system modules quickly after going live with basic enterprise system modules and by continually adding new modules.

Keywords: Economic value of IS, integrated enterprise systems, enterprise resource planning, customer relationship management, supply chain management, implementation strategy

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Introduction

The economic value of enterprise systems has become an evergreen topic in information systems (IS) research (Aral et al., 2012, 2018; Bharadwaj et al., 2007; Hitt et al., 2002; Mithas et al., 2016; Park & Mithas, 2020). IS scholars have investigated the impacts of decision support systems, enterprise resource planning (ERP), customer relationship management (CRM), supply chain management (SCM), artificial intelligence, and data analytics in various contexts (Atasoy et al., 2016; Dhyne et al., 2021; Dong et al., 2009; Havakhor et al., 2019; Mittal & Nault, 2009; Wu et al., 2020). All of these could fall under the broad umbrella of enterprise systems. However, enterprise systems are not isolated because some serve as the foundation for others. For example, internal ERP systems constitute the foundation for internal AI systems and external CRM/SCM systems. Firms usually purchase and implement these systems in an integrated manner, which we call integrated enterprise systems (IES). This integration process often contains the synergy of multiple system modules, IT infrastructure, and business processes (Angst et al., 2011; Benitez et al., 2018; Frohlich & Westbrook, 2001; Nazir & Pinsonneault, 2012; Rai et al., 2015). The existing literature has generated rich findings on the impact of individual systems and compared various returns from different systems (Bharadwaj et al., 1999; Bloom et al., 2014; Melville et al., 2005; Schryen, 2013; Trantopoulos et al., 2017). A few studies have also discussed strategies to implement individual systems and investigated how various implementation strategies influence system performance (Kim & Kankanhalli, 2009; Lee & Xia, 2010; Tan et al., 2020; Thummadi & Lyytinen, 2020), but firms' need and strategy to integrate multiple enterprise systems has received little attention. For firms with multiple enterprise systems to integrate, it is critical that they understand which IES implementation strategies they could apply and how different IES implementation strategies influence the returns from IES. We aim to fill this research gap by examining firms' selection of system modules and decisions on module go-live timing when integrating ERP, CRM, and SCM systems (i.e., the right time to adopt the right modules).

Taking advantage of a unique dataset, we focused on three major suites of enterprise systems, namely, ERP, CRM, and SCM. We combine CRM and SCM as "CRM/SCM" systems because they must be built on ERP to integrate external business processes. Because ERP systems serve as a prerequisite for CRM/SCM systems in terms of data availability and system functionality, firms interested in both systems must integrate them together by going live with them sequentially. We collected enterprise system adoption data on all U.S. customers from the sales database of one large enterprise systems vendor, covering 1997 to 2004.[2] Because existing studies have rarely examined the implementation strategy for the sequential integration of multiple enterprise systems, we borrowed theories from software development literature and enterprise system implementation literature to develop two IES implementation strategies: the agile strategy (simple, quick, and flexible) and the phased strategy (rich, phased, and pre-determined) (Conboy, 2009; Kettunen & Lejeune, 2020; Lee & Xia, 2010; Mahadevan et al., 2015; Thummadi & Lyytinen, 2020). Firms using the agile strategy often go live with CRM/SCM quickly after going live with minimally required ERP modules and then keep adopting additional ERP and/or CRM/SCM modules. In contrast, firms using the phased strategy often go live with CRM/SCM after fully adopting and testing ERP modules. We discuss the rationale for each implementation strategy and demonstrate how each strategy manifests in detailed implementation decisions. Regarding the timing of system adoption and the richness of modules, we focused on three aspects of IES implementation and developed five concepts within these three aspects, including the module foundation of ERP (or CRM/SCM), the sequential connection between ERP and CRM/SCM, and the continued adoption of ERP (or CRM/SCM).

We propose a research framework to explain why and how returns from IES vary with these IES implementation decisions. Our research has three primary goals: (1) to examine the impacts of IES on firm performance, (2) to understand how IES returns vary with different IES implementation decisions, and (3) to understand which strategy, i.e., the agile strategy or the phased strategy, better guides the implementation of IES. We first tested the performance implications of purchasing and going live with ERP systems and CRM/SCM systems. Our data contained distinct entries for purchase and go-live events separately, providing detailed information on module adoption time. Therefore, we were then able to investigate how the returns from IES vary with the agile and phased IES implementation strategies and their related decisions, including the module foundation of ERP (or CRM/SCM), the sequential connection between ERP and CRM/SCM, and the continued adoption of ERP (or CRM/SCM).

[2] This dataset records three types of adoption events: (1) purchase events—firms purchase software packages from the vendor, (2) installation events—firms start to install software packages, and (3) go-live events—firms start to use enterprise systems. We focused on purchase events and go-live events in the main analysis and used installation events in the robustness check.

Our results provide empirical evidence of a positive relationship between IES and firm performance. However, we found that the returns from IES are heterogeneous, depending on the aforementioned IES implementation decisions. Fewer ERP (or CRM/SCM) modules at CRM/SCM go-live events, quicker connection between ERP and CRM/SCM go-live events, and continued adoption of ERP (or CRM/SCM) modules all enhance IES returns. Because these IES implementation decisions are interconnected and collectively reflect the underlying IES implementation strategy, we eventually created a variable measuring the level of agile strategy and found that the agile strategy leads to more IES returns than the phased strategy. Our results are robust to several alternative explanations and specifications, highlighting the importance of the IES implementation strategy in achieving higher returns from IES. Our study suggests that successful firms integrate multiple enterprise systems by going live with SCM/CRM quickly after going live with basic ERP modules and by continually adding new modules. This implication highlights agility and flexibility over readiness and comprehensiveness in the implementation and integration of multiple enterprise systems.

Our findings contribute to the literature by developing and testing the impacts of two IES implementation strategies on IES returns. Our study focuses on the need for firms to sequentially integrate multiple enterprise systems and reveals how a series of IES implementation decisions can help boost returns from IES. To build our theoretical framework, we connected previous studies on the impact of IES and studies on enterprise system implementation (Sykes 2015; Tan et al., 2020). We proposed two competing IES implementation strategies, elaborated on the detailed implementation decisions associated with them, and empirically tested which decisions lead to higher returns from IES. Although studies have documented winning software development strategies and (individual) enterprise system implementation strategies (Mahadevan et al., 2015; Kettunen & Lejeune, 2020; Thummadi & Lyytinen, 2020), the strategy for implementing a successful IES tends to be different. It has been argued that both the agile strategy and the phased strategy are efficient and effective in various contexts, but the agile strategy overwhelmingly outperforms the phased strategy in the context of IES implementation. Our study is among the first to discuss IES implementation strategy and we hope that it motivates future studies to investigate multiple enterprise system implementation as an integrated process.

Theory and Literature

The IS literature has established that enterprise systems can increase firm performance and productivity and that the returns from internal systems and external systems are different (Aral et al., 2012; Berente et al., 2019; Dong et al., 2009; Hendricks et al., 2007; Mithas et al., 2012; Tan et al., 2020). Because we focus on IES, we offer a summary of studies that have investigated the returns from multiple enterprise systems (see Appendix Table A1). For example, Hendricks et al. (2007) found heterogeneous returns from ERP, CRM, and SCM, whereas Bloom et al. (2014) found heterogeneous returns from information technologies (e.g., ERP) and communication technologies (e.g., data intranet). We are more interested in how these systems work together to create value and how firms should implement these systems in a sequence. Because these systems are often not isolated and are likely implemented in an integrated manner, understanding their returns simultaneously is essential.

More importantly, implementing IES requires decisions on several key procedures, e.g., how soon to go live with CRM/SCM after going live with ERP, the number of ERP modules to adopt before going live with CRM/SCM, and the number of CRM/SCM modules to go live with initially. Therefore, the way IES is implemented may determine the returns from IES.[3] By examining these questions, we provide theoretical implications on how IES returns vary with various implementation decisions and offer managerial implications on how to integrate multiple enterprise systems. In a nutshell, we examine when to go live with which modules in the IES implementation process. Angst et al. (2011) documented the importance of the integration sequence in various healthcare systems, which is related to the timing of go-live events in our research. Tian and Xu (2015) studied the risk reduction effect of various enterprise system scopes, which is related to the richness of modules in our research. We advance this literature by considering multiple enterprise systems as integrated systems and focusing on the implementation strategy for IES. Figure 1 shows our theoretical framework, which consists of an illustration of the IES implementation process (top) and a conceptual model of IES implementation strategies (bottom).

[3] We use the term "implementation" to refer to the whole process of system adoption, including purchase events, installment events, and go-live events. We use the term "adoption" to refer to go-live events.

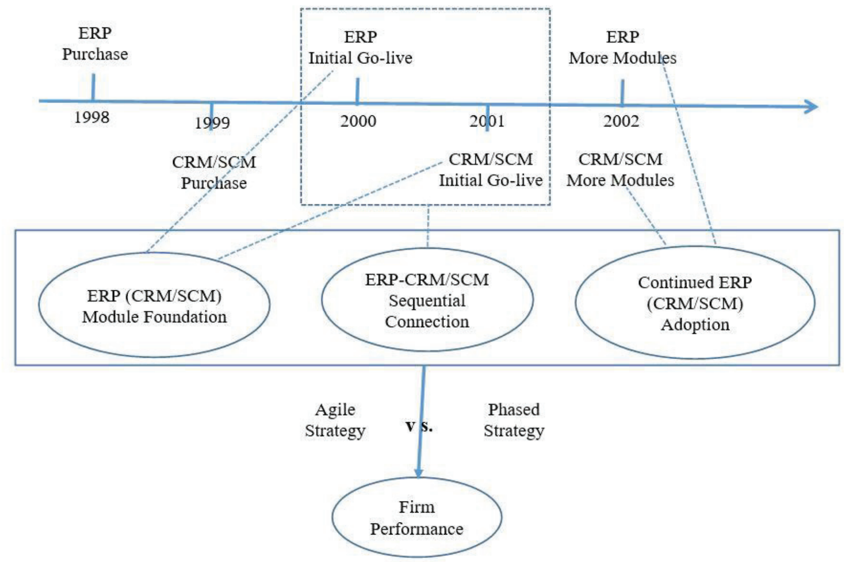

Figure 1. Integrated Enterprise Systems and IES Implementation Strategies

The conceptual model explains how different IES implementation strategies influence the returns from IES. We borrowed ideas from software development literature and enterprise system implementation literature to derive the overarching strategies and related decisions. In software (including information systems) development projects, a common trade-off is between readiness and agility. One idea (as suggested by the waterfall principle) is to follow a static business plan and fully complete and validate each step before moving to the next one, which can eventually lead to a mature and well-designed IT artifact (Kettunen & Lejeune 2020; Mahadevan et al., 2015). For example, the development team must go through requirements, analysis, design, coding, testing, and operations to complete the final product (Thummadi & Lyytinen, 2020). In contrast, the prototyping (or agile) principle advocates a quick and simple version of the product in the initial state, after which a more complex and comprehensive version can be developed based on the experience of the users and designers of the initial version (Kraushaar & Shirland, 1985). Prototyping can shorten development cycles in order to respond to complex, fast-moving, and competitive environments (Ramesh et al., 2010).

A similar trade-off has been documented in the enterprise system implementation literature. One idea (suggested by the phased strategy) is to implement enterprise systems phase by phase because each phase involves distinct activities and different business units (Markus et al., 2000; Ward et al., 2005). When a sequence of systems needs to be implemented, it is even more important to plan each step ahead and fully adopt and test each system before moving to the next. Similarly, the stage-gate strategy delineates a complex process comprising a set of predetermined stages and gates, where stages constitute prescribed tasks, and gates serve as checkpoints at the entrance to each stage (Jagoda & Samaranayake, 2017). This strategy emphasizes readiness and suggests a comprehensive but long process of implementation, including pre-implementation readiness assessment, an implementation plan, an implementation audit, post-implementation assessment, etc. Both the phased strategy and the stage-gate strategy suggest a phased, pre-determined, thoroughly evaluated implementation process, which would hopefully lead to rich functions and few system glitches.

The opposite idea (suggested by the agile strategy) is to adopt the functionality with the highest priority rapidly and let the system run, making adaptive adjustments and refinements later (Armour & Kaisler, 2001). Similarly, the big-bang strategy suggests only adopting modules that match the functionality of the existing systems to avoid possible software modifications and process reengineering (Pliskin & Zarotski, 2000). Both the agile strategy and the big-bang strategy propose that the learning and experimentation experience with the initial basic implementation can help mitigate the risks associated with full-fledged implementation.

Although few studies have directly discussed what implementation strategies firms could use to sequentially integrate multiple enterprise systems, we argue that the common trade-off and primary principles can still be applied in the IES context. Many decisions in the IES implementation face the same trade-off between agility and (full) readiness.

For instance, firms need to decide on when to go live with CRM/SCM after going live with ERP, how many ERP modules to adopt before going live with CRM/SCM, how many CRM/SCM modules to adopt at initial go-live events, and whether to adopt more modules after initial go-live events. Therefore, we use the term “agile strategy” to indicate a simple (fewer initial modules), quick (shorter connection), and flexible (can adopt modules later) implementation approach and use the term “phased strategy” to indicate a rich (more initial modules), phased (longer connection), and pre-determined (adopt all modules as planned before going live) implementation approach. The agile strategy covers the principles of the big-bang strategy and the prototyping strategy in other contexts, whereas the phased strategy covers the principles of the waterfall strategy and stage-gate strategy in other contexts. Based on this distinction, we examine how and why IES returns vary with different IES implementation strategies and decisions.

According to the implementation strategy and the nature of the IES implementation timeline, we derive three theoretical aspects of IES implementation: module foundation, sequential connection, and continued adoption. The module foundation reflects the status of the ERP (CRM/SCM) modules that have been adopted by the time of CRM/SCM go-live events, including the number and domain of ERP (CRM/SCM) modules. It resembles the initial version of a software product, or a system, as defined by the agile and phased strategies. The sequential connection reflects how soon the CRM/SCM project goes live after adopting ERP, which can be as soon as immediately or as long as five years in our data. It resembles the transition/integration speed between two phases or two tasks. Continued adoption reflects whether additional ERP (CRM/SCM) modules are adopted after the initial go-live events of CRM/SCM. It resembles adaptiveness and flexibility, as defined by the agile and phased strategies. We posit that the returns from IES will vary with all these implementation decisions because they determine the IES features and collectively shape the final IES used by the firms. Different from the complementary factors to enterprise systems and/or IT investments, such as corporate governance, R&D spending, business-IT alignment, external IT support, and absorptive capability (Aral et al., 2012; Havakhor et al., 2019; Huang et al., 2022; Mithas & Rust, 2016; Saldanha et al., 2020; Tian & Xu, 2015; Xue et al., 2021), these IES implementation decisions determine the final IES features, which are the central parts and components of IES. Existing studies have documented the connection between IES returns and the final IES features/status according to different aspects, including IT infrastructure flexibility (Benitez et al., 2018), enterprise systems portfolio (Sambhara et al., 2022), IS architecture (Haki et al., 2020), and digital infrastructure (Henfridsson & Bygstad, 2013). Our study extends these studies by identifying the IES implementation decisions that are likely to enable firms to achieve these final features.

To build the baseline in our theoretical framework (Figure 1), we first argue that IES positively influences firm performance (i.e., a positive average treatment effect) because integrating internal and external data is essential to improving supply chain coordination and mitigating firm risk (Frohlich & Westbrook 2001; Tian & Xu, 2005). In particular, ERP systems provide integrated data and interfaces that enable effective execution of supply chain activities and utilization of customer data. Information inputs from supply chain partners can directly automate the planning of internal production activities and help optimize distribution and logistics (Yao & Zhu, 2012). For our theory-building purpose, we aim to first replicate the well-known positive impact of IES on firm performance. Second, we propose that all three aspects of IES implementation, including module foundation, sequential connection, and continued adoption, shape the final IES differently and influence the returns from the final IES. We derive five related concepts from these three aspects. Because existing studies rarely discuss IES implementation strategies and decisions, we adapt core principles and common arguments from the agile and phased strategies in other contexts to develop our theoretical expectations. We summarize these concepts and their impacts on the returns from IES in Table 1.

ERP module foundation is expected to influence the returns from IES because CRM/SCM systems take advantage of ERP systems to further integrate external processes. However, it is unclear how many modules are sufficient to support a successful IES. The agile strategy always seeks to avoid prescribing cumbersome and time-consuming processes and tasks that add little value to the software product or system implementation (Fitzgerald et al., 2006). It tends to prioritize the swift implementation of IES, suggesting that a few ERP modules that can satisfy the minimum requirement would be sufficient to go live with CRM/SCM. This approach is likely to be more feasible technically, cause less resistance, and allow the whole IES to take effect faster and sooner. In contrast, the phased strategy values readiness and preparedness, suggesting that firms should go live with rich ERP modules before going live with CRM/SCM, under the assumption that richer ERP modules would better facilitate CRM/SCM. Due to the opposite arguments of the two strategies, we don’t have a clear prior hypothesis on the direction of the impact of ERP module foundation on IES returns. The same logic also applies to CRM/SCM module foundation. The agile strategy suggests that firms should start IES quickly and adjust later, whereas the phased strategy advocates planning ahead and adopting all desired modules at one time. As a result, the agile strategy suggests that basic CRM/SCM modules should be adopted at the initial go-live events, whereas the phased strategy suggests using rich CRM/SCM modules as a foundation.

Table 1. IES Implementation Strategies and Decisions

Decision-related concepts	Implementation questions	Expectations from the agile strategy	Expectations from the phased strategy	Measures
ERP module foundation	What number and domains of ERP modules should be adopted before going live with CRM/SCM?	Basic modules are good enough to start CRM/SCM because a workable system is more important.	Rich modules have to be adopted before starting CRM/SCM because a solid foundation is important.	(1) Number of ERP modules (2) Domain of ERP modules
ERP-CRM/SCM sequential connection	How soon should firms go live with CRM/SCM after going live with ERP?	Immediately if possible because they must work together and be tested together.	After a while because firms must make sure ERP works properly first.	Whether there is a gap between the ERP go-live event and the CRM/SCM go-live event
CRM/SCM module foundation	What number and domains of CRM/SCM modules should be adopted at the time of CRM/SCM go-live events?	Basic modules are good enough to start CRM/SCM because a workable system is more important.	Rich modules have to be adopted because rich modules provide rich functions.	1) Number of CRM/SCM modules (2) Domain of CRM/SCM modules
Continued ERP adoption	Should firms go live with additional ERP modules after going live with CRM/SCM?	Yes, because firms have to keep monitoring and improving IES.	No need because firms should have already tested and fixed all ERP issues before going live with CRM/SCM.	Whether additional ERP modules were adopted after CRM/SCM go-live events
Continued CRM/SCM adoption	Should firms go live with additional CRM/SCM modules after going live with initial CRM/SCM modules?	Yes, because firms must keep monitoring and improving IES.	No need because firms should have already planned and adopted all desired CRM/SCM modules initially.	Whether additional CRM/SCM modules were adopted after CRM/SCM go-live events

Another critical concept is the connection between ERP and CRM/SCM. Because the adoption of CRM/SCM goes across firm boundaries, it requires both internal and external process improvement and redesign. The agile strategy advocates a quick (if not immediate) connection, meaning that firms can run and test IES as a whole, which may help firms and their partners quickly gain IES maintenance and coordination experience. This seamless connection has the capacity to coordinate the internal and external process redesign simultaneously. The phased strategy suggests that the implementation and transaction processes should take time, which could facilitate the testing of ERP stability and the user assimilation of ERP-related knowledge (Robey et al., 2002). Therefore, the phased strategy suggests that firms should not rush to adopt CRM/SCM after going live with ERP. Due to the opposing arguments of the two strategies, we do not have a clear prior hypothesis on the direction of the impact of ERP-CRM/SCM sequential connection on IES returns.

Continued ERP adoption may influence IES returns by complementing the existing IES by providing additional functionalities or business data. The agile strategy promotes the power of feedback from users and adaptive system adoption (Armour & Kaisler, 2001; Kraushaar & Shirland, 1985; Maruping & Matook, 2020). Thus, with this approach, it is not uncommon to see new modules adopted after initial go-live events. If this rearrangement of module adoption sequence works, we would expect firms that go live with additional ERP modules after going live with CRM/SCM to see higher returns from IES. The phased strategy, in contrast, would follow a suite-by-suite approach (i.e., ERP suite → CRM/SCM suite). Using this approach, it is unlikely that additional ERP modules would be adopted after going live with CRM/SCM. If it is indeed beneficial for firms to go live with all required/planned modules at the same time to synergize their functions and business data, we would expect firms that do not go live with additional ERP modules after going live with CRM/SCM to see higher returns from IES. Due to the opposing arguments of the two strategies, we do not have a clear prior hypothesis on the direction of the impact of continued ERP adoption on IES returns. The same logic also applies to continued CRM/SCM adoption, which may increase IES returns according to the agile strategy but decrease IES returns according to the phased strategy.

Empirical Methods

We collected detailed data on the enterprise system purchase and go-live decisions of 2,132 firms from 1997 to 2004, 675 of which were publicly traded and had matched performance data in the Compustat database. We thus created an annual panel of 675 firms over eight years. The data include all U.S. sales of a major vendor's 150 software modules sold during the study period. These modules represent major packaged software offerings, including ERP and CRM/SCM. The adoption of ERP and CRM/SCM typically spans several years punctuated by discrete observable events, including the purchase date of a system module, the installment date, and the go-live date. As a result, we could observe the exact time of IES implementation on multiple system models and the richness of modules in each go-live event. This information enabled us to identify the IES implementation strategy used by firms and the key implementation decisions they made, eventually helping us estimate IES returns under each implementation decision.

One of the most vexing problems in estimating the performance impacts of IT is simultaneity bias, that is, errors introduced into estimations of variables simultaneously determined by the same forces (Olley & Pakes, 1996). If positive shocks to firm performance occur at particular times during the observation period, they may simultaneously affect investments in IT and the performance of firms. We handle this challenge by separately estimating IES purchases and IES use (Aral et al., 2012). Because we can observe the exact dates of purchase and go-live events, we examine the impact of IES adoption on firm performance by explicitly controlling for the association between IES investment and firm performance. This advantage helped us disentangle the causal direction of the relationship between IES adoption and performance. If firm performance is correlated with use but uncorrelated with purchase, we can reasonably assume the relationship with performance is not being driven (at least primarily) by the simultaneous determination of investment and performance.

Following the literature on IT value, we employed a log-log specification to identify the impact of IES (Aral et al., 2012; Hitt et al., 2002). In line with previous studies, the specification uses return on assets (ROA) and return on equity (ROE) to measure firm performance (Dewan & Ren, 2011; Kohli et al., 2012). ROA is defined by pretax income divided by assets, with larger values indicating more efficient operation of a firm. ROE is measured by pretax income divided by equity, with larger values indicating higher returns accruing to the common shareholders. We arrived at the following equation as our main specification to study the relationship between IES and firm performance:

$$lnPerformanceLHS_{it} = \alpha_{it} + \beta_1 lnPerformanceRHS_{it} + \beta_2 Purchase_{it} + \beta_3 GoLive_{it} + YearFixedEffects_t \times Industry_i + FirmFixedEffects_i + \varepsilon_{it} \quad (1)$$

ln$PerformanceLHS_{it}$ is the natural log transformation of the performance outcome variable, and ln$PerformanceRHS_{it}$ is the natural log transformation of the performance control variable.[4] The $Purchase_{it}$ and $GoLive_{it}$ variables for firm i and year t can be applied to any specific enterprise system. In the main model, we mark $Purchase_{it}$ as 1 in and after the year of the purchase event, and 0 otherwise. The same logic applies to $GoLive_{it}$. This coding approach makes Equation (1) a difference-in-differences model (Bertrand et al., 2004; Mithas et al., 2022). The estimation of the returns from IES thus mainly comes from the variation in ERP and CRM/SCM adoption dates. B_3 is the coefficient of interest, estimating the impact of IES on the change ratio of pretax income when controlling for the change ratio of total assets (or equity). To control for the unobserved time-invariant firm characteristics and time-variant industry-specific trends, we also included firm fixed effects and year-industry fixed effects (see Appendix Table B3 for coding details on industry category).

To examine the heterogeneous returns from IES (Aral et al., 2012; Tian & Xu, 2015), we added an interaction term of the $GoLive_{it}$ variable and the IES implementation decisions to Equation (1). The IES implementation decisions include the ERP (CRM/SCM) module foundation, ERP-CRM/SCM sequential connection, and continued ERP (CRM/SCM) adoption. Equation (2) shows the empirical specification:

$$lnPerformanceLHS_{it} = \alpha_{it} + \beta_1 lnPerformanceRHS_{it} + \beta_2 Purchase_{it} + \beta_3 GoLive_{it} + \beta_4 GoLive_{it} \times IESImplementationDecisions_i + YearFixedEffects_t \times Industry_i + FirmFixedEffects_i + \varepsilon_{it} \quad (2)$$

Table 2 reports the descriptive statistics of the major variables used in this study. We will elaborate on the definitions and measurements of these variables when we use them.[5]

[4] One concern of the log-log specification is that it excludes firms that have negative pretax income, which is probably fine because firms with negative pretax income tend to have extreme ROA (or ROE) in our data and do not represent typical ERP (or CRM/SCM) adopters. The impact of IES on these firms may not be reflected by financial performance but by manufacturing capacity or user base.

[5] Based on the definitions and measurements of the three panels of variables, we report the correlation of Panel A and Panel B variables in Appendix Table B1 and the correlation of Panel C variables in Appendix Table B2.

Table 2. Descriptive Statistics						
Variable	**Obs.**	**Mean**	**Median**	**S.D.**	**Min**	**Max**
Panel A: Performance variables						
Total assets (in millions)	4,612	12,479	1,592	39,105	0.07	801,145
Total equity (in millions)	4,401	3,616	662	9,009	0.008	152,027
Pretax income (in millions)	4,602	381	40	1,812	-44,574	25,330
Panel B: Enterprise systems adoption features						
ERP purchase status	5,400	0.658	1	0.475	0	1
ERP live status	5,400	0.366	0	0.482	0	1
CRM/SCM purchase status	5,400	0.370	0	0.483	0	1
CRM/SCM live status	5,400	0.142	0	0.349	0	1
Panel C: IES implementation decisions						
Number of ERP modules	211	4.934	4	3.898	1	16
No ERP-CRM/SCM going-live gap	211	0.597	1	0.492	0	1
Number of CRM/SCM modules	211	1.872	1	1.283	1	7
Continued ERP adoption	211	0.137	0	0.345	0	1
Continued CRM/SCM adoption	211	0.289	0	0.454	0	1
Agile strategy level	211	2.322	2	1.104	0	5

Note: All IES implementation decisions (Panel C) are firm specific, time invariant, and only meaningful for firms that have gone live with CRM/SCM. Our data contained 675 firms, among which 675 firms purchased ERP, 422 firms went live with ERP, 452 firms purchased CRM/SCM, and 211 firms went live with CRM/SCM. Because some firms may go public or change status during our observation period and some performance measures are not always available, the number of observations does not always equal 5,400 (i.e., 675 × 8).

Results

Returns from Integrated Enterprise Systems: CRM/SCM

As mentioned in the Theory and Literature section, we expect IES to have a positive impact on firm performance. In our context, firms must adopt ERP first and then CRM/SCM to integrate them. Because firms adopt them sequentially, we estimate them simultaneously to determine whether the adoption of IES (rather than ERP only) drives performance. Therefore, we distinguish between the go-live events of ERP and CRM/SCM. The go-live event of CRM/SCM is defined as that of either the CRM or SCM module, whichever comes first for a particular firm. To control for the potential confounding effect of simultaneity or reverse causality, we also included purchase events in the model. We applied two approaches to control for purchase events due to two alternative assumptions.[6]

Table 3 reports the results and shows that CRM/SCM adoption has a positive and significant impact on firm performance. The impacts are around 11% for both ROA and ROE. In other words, the adoption of CRM/SCM leads to an 11% increase in pretax income, controlling for the change rate of total assets or equity. None of the purchase events are significantly related to firm performance, ruling out the explanation of reverse causality (i.e., good performance leads to IT adoption). This finding is robust to a battery of robustness checks (see the Robustness Check section), including lagged effects of ERP, omitted variable bias, alternative performance measures, alternative CRM/SCM coding, and so on. The impact of ERP is negative and sometimes significant, suggesting the mere adoption of ERP does not increase firm performance. Only when firms adopt the full suites of ERP and CRM/SCM (i.e., IES) do firms see higher returns. The combined impact of ERP and CRM/SCM is positive but insignificant, which is due to the heterogeneous returns from IES and highlights the importance of proper IES implementation strategies and decisions (we will present more evidence later).

How Do the Returns from IES Vary with ERP Module Foundation?

Since ERP serves as the foundation for CRM/SCM, understanding the impact of ERP module foundation is important for a successful IES implementation. We focused on two aspects: the number of ERP modules and the domain of ERP modules.

[6] If purchase was driven by long-lasting firm strategic initiatives, we focused on purchase status and measured it in the same way as live status (i.e., purchase status variable taking a value of 1 in and after the initial purchase year). If purchase was driven by a one-time firm decision, we focused on purchase action and marked the corresponding variable as 1 only when a purchase happened in that particular year (i.e., purchase-action variable taking a value of 1 in purchase years only). Because the two approaches returned almost identical results, we applied the first approach in later analyses only.

Table 3. Returns from ERP and CRM/SCM: During Purchase and after Going Live

	(1)	(2)	(3)	(4)
Dependent variable	ln(pretax income)	ln(pretax income)	ln(pretax income)	ln(pretax income)
Interpretation	ROA	ROA	ROE	ROE
ln(total assets)	0.765^{***} (0.059)	0.763^{***} (0.059)		
ln(total equity)			0.604^{***} (0.049)	0.603^{***} (0.049)
ERP purchase status	-0.017 (0.086)		-0.024 (0.081)	
ERP purchase action		0.032 (0.068)		0.037 (0.060)
ERP live status	-0.060 (0.045)	-0.067 (0.052)	-0.091^{**} (0.033)	-0.097^{**} (0.042)
CRM/SCM purchase status	-0.024 (0.047)		0.001 (0.048)	
CRM/SCM purchase action		-0.062 (0.074)		-0.057 (0.066)
CRM/SCM live status	0.108^{***} (0.029)	0.104^{**} (0.042)	0.111^{**} (0.036)	0.113^{*} (0.052)
Other control variables	Firm year × Industry	Firm year × Industry	Firm year × Industry	Firm year × Industry
Adjusted R^2	0.88	0.88	0.89	0.89
Observations	3,336	3,336	3,280	3,280

Note: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Standard errors are clustered at the industry level to control for industry-specific trends/dynamics/competitions. The number of observations varies due to the availability of certain performance measures. The coefficients are close to significant (e.g., p-value <0.15) when we cluster standard errors at the firm level (see Appendix C Table C2); thus, please use caution in interpreting the average treatment effect.

Table 4. Returns from CRM/SCM: ERP Module Foundation

	(1)	(2)	(3)	(4)
Dependent variable	ln(pretax income)	ln(pretax income)	ln(pretax income)	ln(pretax income)
Interpretation	ROA	ROA	ROE	ROE
CRM/SCM purchase status	-0.028 (0.048)	-0.025 (0.047)	-0.003 (0.049)	-0.000 (0.047)
CRM/SCM live status	0.271^{***} (0.061)		0.273^{***} (0.058)	
CRM/SCM live status × # of ERP modules	-0.030^{**} (0.010)		-0.029^{***} (0.008)	
CRM/SCM live status × ERP module domains—basic or two		0.296^{***} (0.077)		0.296^{***} (0.073)
CRM/SCM live status × ERP module domains—three		0.000 (0.016)		0.017 (0.090)
CRM/SCM live status × ERP module domains—four		-0.022 (0.084)		-0.019 (0.082)
Other control variables	√	√	√	√
Adjusted R^2	0.88	0.88	0.89	0.89
Observations	3,336	3,336	3,280	3,280

Note: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Standard errors are clustered at the industry level.

Because ERP is made up of dozens of functional modules, how many modules are required to launch a successful CRM/SCM system is unclear. We focused on the number of modules each firm had when it went live with CRM/SCM as the first indicator of ERP module foundation. In Columns 1 and 3 of Table 4, we report the results, showing that the interaction term "CRM/SCM live status × # of ERP modules" is negative and significant. This result suggests that if firms went live with more ERP modules before going live with CRM/SCM, it actually led to lower returns from CRM/SCM.[7]

Because the number of modules may mask the effect of different modules, we also grouped ERP modules into four domains and investigated how the different domains of ERP modules influence the returns from CRM/SCM. The four

[7] We used the term "the returns from IES" to refer to how the full suite of ERP and CRM/SCM influences firm performance. Because CRM/SCM has to be built upon ERP and an IES can be considered as complete only after a firm adopted CRM/SCM, we focused on "the returns from CRM/SCM" and treated "the returns from ERP" as the common baseline. In other words, when we compared the heterogeneous returns from IES, we only needed to compare the heterogeneous returns from CRM/SCM. We assumed the returns from ERP are not influenced by IES implementation strategies and decisions because these actions (e.g., ERP-CRM/SCM sequential connection) happened after ERP go-live events and did not have to exist for ERP to take effect. In other words, we assumed the returns from ERP are homogenous to IES implementation strategies and decisions. As a result, in our empirical analysis, the coefficient of "CRM/SCM live status" naturally reflects the returns from CRM/SCM and we can interpret it as or connect it directly to the returns from IES.

ERP module domains are the basic module domain, the human resources module domain, the finance module domain, and the internal operations module domain. All CRM/SCM adopters went live with the basic module domain (because this is the prerequisite to go live with CRM/SCM), but the other domains are optional and firms may have gone live with none of them, one of them, two of them, or all three of them. Based on the number of domains that firms went live with, we created three mutually exclusive dummy variables to reflect firms' module domain decisions: "ERP module domains—basic or two" takes the value of 1 if a firm went live at most two ERP domains when it went live with CRM/SCM; "ERP module domains—three" takes the value of 1 if a firm went live with three ERP domains; "ERP module domains—four" takes the value of 1 if a firm went live all four ERP domains. The interaction terms between these ERP module domain dummy variables and "CRM/SCM live status" reflect the impacts of CRM/SCM under different ERP module domain decisions. The results in Columns 2 and 4 once again suggest that fewer module domains lead to higher returns from CRM/SCM. One explanation could be that the basic module domain (or with one more additional domain) can enable firms to focus on the key processes and fundamental functions of ERP, avoiding the coordination and maintenance costs associated with multiple module domains. To summarize, our results favor the agile strategy over the phased strategy of ERP module foundation (either measured by the total number of modules or of domains). One remaining concern could be the heterogeneous impacts of various module domains, which we address in Appendix D.

How Do the Returns from IES Vary with ERP-CRM/SCM Sequential Connection?

One essential yet understudied question in the IES context is how the sequential connection between ERP and CRM/SCM influences IES returns. For example, how soon should the firms go live with CRM/SCM after going live with ERP? To answer this question, we created a new variable, "going-live gap," measuring the year difference between the CRM/SCM go-live event and ERP go-live event. Although the going-live gap ranges between 0 and 5, around half of the CRM/SCM adopters had no going-live gap (i.e., these firms went live with ERP and CRM/SCM in the same year). Therefore, we created the dummy variable "no going-live gap" to indicate whether the going-live gap is 0 or not. Columns 1 and 3 of Table 5 report the results on the sequential connection of IES. The interaction term "no going-live gap" and "CRM/SCM live status" estimates the returns from CRM/SCM when firms went live with ERP and CRM/SCM in the same year, while the interaction term "with going-live gap" and "CRM/SCM live status" estimates the returns from CRM/SCM when firms went live with CRM/SCM at least one year after going live with ERP. The results suggest that a swift connection (i.e., no going-live gap) is essential for firms seeking to benefit from IES. Longer connection times damage the value of CRM/SCM, probably due to the separation between ERP and CRM/SCM or the inability to incorporate real-time user feedback into IES. A potential concern of this analysis is that the going-live gap is related to the time of CRM/SCM go-live events; thus, we excluded CRM/SCM adopters that went live with CRM/SCM in or after 2002 (which is the median cutoff of the CRM/SCM go-live year) as a robustness check. This approach revealed similar results (see Columns 2 and 4), ruling out the alternative explanation that there was not enough time for CRM/SCM to take effect.

How Do the Returns from IES Vary with the CRM/SCM Module Foundation?

In Table 3, we show that the average treatment effect of CRM/SCM is positive and significant, but in reality, the firms that went live with CRM/SCM adopted different CRM/SCM modules and thus may not have necessarily seen the same returns from CRM/SCM. In this section, we examine how the number and the various domains of CRM/SCM modules influence the impact of CRM/SCM. The influence could be positive because the integration and synergy of multiple modules may support each other and facilitate external coordination and cooperation. However, the influence could also be negative because integrating multiple modules is complex and requires wide involvement and significant time and coordination costs. We first measured CRM/SCM module foundation by the total number of CRM/SCM modules a firm went live with at the initial go-live event (e.g., if a firm initially went live with three CRM/SCM modules at the same time, the value of this variable is three). Like the previous specification used in the analysis of ERP module foundation, the variable of interest is "CRM/SCM live status × # of CRM/SCM module," which represents the additional effect of the number of CRM/SCM modules. Columns 1 and 3 of Table 6 report the results, which suggest a negative effect associated with the number of modules. Both coefficients are negative, and the one in Column 1 is also significant (the one in Column 3 is not significant but the magnitude of impact is similar). The results suggest that when firms go live with CRM/SCM with more modules, the returns from CRM/SCM become smaller and may be negative if there are five or more modules (i.e., 0.182/0.045 = 4.04 and 0.168/0.036 = 4.67).

Table 5. Returns from CRM/SCM: ERP-CRM/SCM Sequential Connection				
	(1)	(2)	(3)	(4)
Dependent variable	ln(pretax income)	ln(pretax income)	ln(pretax income)	ln(pretax income)
Interpretation	ROA	ROA	ROE	ROE
Sample	Full	No late adopters	Full	No late adopters
CRM/SCM purchase status	-0.022 (0.047)	-0.014 (0.038)	0.002 (0.047)	0.007 (0.040)
CRM/SCM live status × No going-live gap	0.201^{***} (0.045)	0.271^{***} (0.073)	0.227^{***} (0.053)	0.319^{***} (0.059)
CRM/SCM live status × With going-live gap	0.011 (0.089)	-0.059 (0.176)	-0.010 (0.101)	-0.035 (0.208)
Other control variables	√	√	√	√
Adjusted R^2	0.88	0.88	0.89	0.89
Observations	3,336	2,751	3,280	2,706

Note: ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$. Standard errors are clustered at the industry level. "No late adopters" means we excluded CRM/SCM adopters that went live with CRM/SCM in or after 2002 from the full sample.

Table 6. Returns from CRM/SCM: CRM/SCM Module Foundation				
	(1)	(2)	(3)	(4)
Dependent variable	ln(pretax income)	ln(pretax income)	ln(pretax income)	ln(pretax income)
Interpretation	ROA	ROA	ROE	ROE
CRM/SCM purchase status	-0.027 (0.047)	-0.028 (0.047)	-0.001 (0.048)	-0.002 (0.047)
CRM/SCM live status	0.182^{***} (0.051)		0.168^{**} (0.068)	
CRM/SCM live status × # of CRM/SCM modules	-0.045^{*} (0.021)		-0.036 (0.024)	
CRM/SCM live status × CRM/SCM module domains—data		0.105^{**} (0.037)		0.116^{*} (0.059)
CRM/SCM live status × CRM/SCM module domains—operation		0.159^{a} (0.088)		0.139^{*} (0.073)
CRM/SCM live status × CRM/SCM module domains—data and operation		0.032 (0.067)		0.045 (0.071)
Other control variables	√	√	√	√
Adjusted R^2	0.88	0.88	0.89	0.89
Observations	3,336	3,336	3,280	3,280

Note: ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$. Standard errors are clustered at the industry level. A: p-value = 0.101.

We also distinguished the functional domains of various CRM/SCM modules and grouped them into data domains (e.g., data warehouse module) and operation domains (e.g., B2B commerce module, supply chain planner, and optimizer module). All firms adopted at least one domain and some firms adopted both. Based on this fact, we created three mutually exclusive dummy variables to indicate the domain foundation of CRM/SCM modules. "CRM/SCM module domains—data" means that the firm adopted data-related modules only; "CRM/SCM module domains—operation" means that the firm adopted operation-related modules only; "CRM/SCM module domains—data and operation" means the firm adopted both data-related and operation-related modules. The findings in Columns 2 and 4 suggest that firms adopting one module domain have more gains than firms adopting two module domains. This finding is consistent with the finding that more CRM/SCM modules at the initial go-live event led to lower returns. Overall, the analysis of CRM/SCM module foundation indicates that the coordination and the business reengineering costs of multiple CRM/SCM modules outweigh the synergetic benefit of a rich module foundation, leading to fewer performance gains.

How Do the Returns from IES Vary with Continued ERP Adoption?

As discussed in the theory-building section, one option for IES implementation, advocated by the phased strategy, is to go live with CRM/SCM along with all needed and planned ERP modules. In contrast, the agile strategy suggests an alternative option—adopting the minimum number ERP modules before CRM/SCM go-live events and continuing to adopt new ERP modules after that. To distinguish these two decisions, we created a dummy variable called "continued ERP adoption," which measured whether firms adopted additional ERP modules after the year in which they went live with CRM/SCM. Continued adoption explicitly reflects a dynamic and flexible implementation process because firms are likely to keep monitoring the IES and add new modules accordingly. We interacted "continued ERP adoption"/ "no continued ERP adoption" with "CRM/SCM live status" to examine how the returns from CRM/SCM vary if firms continue to adopt ERP. Columns 1 and 3 of Table 7 report the results and suggest that although firms that do not continue ERP adoption after going live with CRM/SCM can see sizeable (though insignificant) returns from CRM/SCM, firms that keep adding ERP modules even after going live with CRM/SCM see much higher returns from CRM/SCM. The results highlight the importance of continued IES implementation. After firms gain some experience in coordinating ERP and CRM/SCM and receive feedback from key users, they are more likely to complement the IES with the needed modules. As a robustness check, we excluded late CRM/SCM adopters from the sample to make sure all adopters were likely to have the option to adopt new ERP modules within our observation period. The results in Columns 2 and 4 are largely consistent.

How Do the Returns from IES Vary with Continued CRM/SCM Adoption?

Like the option of continued ERP adoption, firms need to decide whether to adopt more CRM/SCM modules after going live with the first batch of CRM/SCM modules. We created a dummy variable, "continued CRM/SCM adoption," to reflect this option. It takes the value of 1 if a firm adopted additional CRM/SCM modules after it went live with the first batch of CRM/SCM modules. We focused on the interaction terms between "continued CRM/SCM adoption"/ "no continued CRM/SCM adoption" and "CRM/SCM live status." Columns 1 and 3 of Table 8 report the results and suggest that although firms that don't continue CRM/SCM adoption after going live with CRM/SCM can see sizeable (though insignificant) returns from CRM/SCM, firms that keep adding CRM/SCM modules even after going live with CRM/SCM see much higher returns. The findings are similar to the ones on continued ERP adoption, and the reasons are also similar: firms that keep monitoring and building on the current IES receive higher returns. We conducted a robustness check and report the results in Columns 2 and 4 of Table 7. The performance difference between the two types of CRM/SCM adopters becomes larger in the subsample, indicating an even stronger impact of continued CRM/SCM adoption.

How Do the Returns from IES Vary between The Agile Strategy and the Phased Strategy?

In the previous analysis, we focused on every single decision of IES implementation because it is simple to group firms based on one IES implementation decision at one time. However, these decisions are usually interconnected and collectively reflect the underlying IES implementation strategy. An agile strategy can be reflected by fewer domains of ERP modules at CRM/SCM go-live events (i.e., the second measurement of ERP module foundation), no gap between ERP and CRM/SCM go-live events (i.e., the measurement of ERP-CRM/SCM sequential connection), fewer domains of CRM/SCM modules at CRM/SCM go-live events (i.e., the second measurement of CRM/SCM module foundation), continued ERP adoption, and continued CRM/SCM adoption. Based on the values of these variables, we assigned an agile strategy level to each firm (see Appendix D for more details). This level ranges from 0 to 5 with a median value of 2 and a mean value of 2.32 (only a few firms are able to strictly follow the practice of the agile strategy or the phased strategy, as suggested by Fitzgerald et al., 2006, and Thummadi and Lyytinen, 2020). We interacted "agile strategy level" with "CRM/SCM live status" and interpreted the coefficient of this interaction term as the performance gains associated with the practice of the agile strategy. The results in Columns 1 and 3 of Table 9 suggest that the agile strategy boosts the impact of IES more than the phased strategy because higher agile strategy levels lead to higher returns from CRM/SCM. Based on the spectrum of the agile strategy level, we also grouped firms into three agile strategy categories to better distinguish the impacts of various IES implementation strategies, i.e., the "phased strategy" (agile strategy level = 0 or 1), the "semi-agile strategy" (agile strategy level = 2 or 3), and the "agile strategy" (agile strategy level = 4 or 5). We then investigated the returns from CRM/SCM based on this classification. The results in Columns 2 and 4 of Table 9 suggest that firms using the semi-agile strategy or agile strategy benefit from CRM/SCM, but firms using the phased strategy do not see positive returns from CRM/SCM.

The findings are consistent with previous analyses and overwhelmingly favor the agile strategy over the phased strategy. The high implementation and coordination costs of rich modules under the phased strategy are likely to hinder the synergy of ERP and CRM/SCM models, making it hard for firms to take full advantage of IES functions. In contrast, by using the agile strategy, firms can focus on the minimum required modules to quickly complete process reengineering and synergize ERP with CRM/SCM. This strategy guarantees that an initial version of IES will work, which helps to reduce resistance and gain feedback to prepare for continued module adoption and function extension.

Table 7. Returns from CRM/SCM: Continued ERP Adoption

	(1)	(2)	(3)	(4)
Dependent variable	ln(pretax income)	ln(pretax income)	ln(pretax income)	ln(pretax income)
Interpretation	ROA	ROA	ROE	ROE
Sample	Full	No late adopters	Full	No late adopters
CRM/SCM purchase status	-0.024 (0.048)	-0.020 (0.065)	0.000 (0.060)	0.001 (0.061)
CRM/SCM live status × No continued ERP adoption	0.079* (0.038)	0.083 (0.103)	0.082 (0.073)	0.113 (0.104)
CRM/SCM live status × With continued ERP adoption	0.298* (0.160)	0.373* (0.210)	0.298* (0.179)	0.435** (0.213)
Other control variables	√	√	√	√
Adjusted R^2	0.88	0.88	0.89	0.89
Observations	3,336	2,751	3,280	2,706

Note: ****p* < 0.01; ***p* < 0.05; **p* < 0.1. Standard errors are clustered at the industry level. "No late adopters" means we excluded CRM/SCM adopters that went live with CRM/SCM in or after 2002 from the full sample.

Table 8. Returns from CRM/SCM: Continued CRM/SCM Adoption

	(1)	(2)	(3)	(4)
Dependent variable	ln(pretax income)	ln(pretax income)	ln(pretax income)	ln(pretax income)
Interpretation	ROA	ROA	ROE	ROE
Sample	Full	No late adopters	Full	No late adopters
CRM/SCM purchase status	-0.024 (0.047)	-0.019 (0.038)	0.000 (0.048)	0.001 (0.041)
CRM/SCM live status × No continued CRM/SCM adoption	0.074 (0.061)	0.086 (0.106)	0.066 (0.065)	0.092 (0.120)
CRM/SCM live status × With continued CRM/SCM adoption	0.181* (0.083)	0.207*** (0.062)	0.207** (0.087)	0.290*** (0.048)
Other control variables	√	√	√	√
Adjusted R^2	0.88	0.88	0.89	0.89
Observations	3,336	2,751	3,280	2,706

Note: ****p* < 0.01; ***p* < 0.05; **p* < 0.1. Standard errors are clustered at the industry level. "No late adopters" means we excluded CRM/SCM adopters that went live with CRM/SCM in or after 2002 from the full sample.

Table 9. Returns from CRM/SCM: The Agile Strategy vs. The Phased Strategy

	(1)	(2)	(3)	(4)
Dependent variable	ln(pretax income)	ln(pretax income)	ln(pretax income)	ln(pretax income)
Interpretation	ROA	ROA	ROE	ROE
CRM/SCM purchase status	-0.026 (0.047)	-0.026 (0.048)	-0.002 (0.048)	-0.002 (0.048)
CRM/SCM live status	-0.239 (0.161)		-0.261 (0.163)	
CRM/SCM live status × Agile strategy level	0.160** (0.069)		0.171** (0.070)	
CRM/SCM live status × Phased strategy		-0.130 (0.148)		-0.142 (0.132)
CRM/SCM live status × Semi-agile strategy		0.171* (0.085)		0.177** (0.073)
CRM/SCM live status × Agile strategy		0.495** (0.217)		0.545** (0.229)
Other control variables	√	√	√	√
Adjusted R^2	0.88	0.88	0.89	0.89
Observations	3,336	3,336	3,280	3,280

Note: ****p* < 0.01; ***p* < 0.05; **p* < 0.1. Standard errors are clustered at the industry level. The results were similar when we just used the agile strategy and the phased strategy to cover all agile strategy levels.

Robustness Checks

To validate our findings, we considered multiple alternative explanations and ruled out many of them empirically. We also conducted conventional robustness checks specific to the difference-in-differences method. Appendix C summarizes various concerns, our corresponding empirical solutions, and results. Our main findings continue to hold.

Discussion and Conclusion

How and why IES causes performance increase is a critical question in IS research. In this paper, we used data at the module level to illuminate how different IES implementation strategies and decisions influence the returns from IES. We distinguished between the agile strategy and the phased strategy, as well as their associated IES implementation decisions. Our results show a positive impact of CRM/SCM on firm performance and confirm the critical role of IES implementation decisions in influencing IES returns. Our findings favor the agile strategy over the phased strategy, suggesting that firms should focus on the minimum required ERP and CRM/SCM modules and prioritize swift integration over a comprehensive but slow integration. It is more beneficial to keep adopting additional modules after the initial adoption than it is to go live with all modules simultaneously.

Our findings provide direct implications on how firms should sequentially integrate multiple enterprise systems. First, the faster the implementation is, the more gains firms can obtain. A comprehensive ERP module foundation might sound good for IES, but we found this to be untrue, probably because the long implementation of ERP may mask the importance of integrating and coordinating ERP and CRM/SCM. Second, firms do not need to adopt all ERP modules before going live with CRM/SCM systems. Even if they only adopt a few key ERP modules, firms can still benefit from CRM/SCM. This implication also applies to CRM/SCM modules: fewer CRM/SCM modules (rather than richer CRM/SCM modules) lead to higher returns from IES. Finally, it is essential for firms to keep monitoring the IES and adding new modules, which may complement or strengthen the functionality of the current system. It is worth noticing that the agile implementation strategy is not free of costs. For example, in the information systems development context, Benlian (2022) documented that agile practices may hurt developers.

Our paper is not without limitations. While we documented how the agile and phased implementation strategies and decisions influence the returns from IES, we were unable to identify what factors motivate firms to select a certain strategy and how internal capabilities and external environments moderate the impact of each implementation strategy. For example, the phased strategy has been documented as effective in some cases. Future studies could explore the alignment between the IES implementation strategy and firms' objectives and capabilities.

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Appendix A

Appendix A summarizes the studies on IES returns and the factors that influence the returns from IES. Although existing studies have investigated the returns from IES and identified a variety of external factors that influence the IES returns, we are among the first to examine how IES implementation strategies and decisions impact the returns from IES.

Table A1. Literature Review of Studies on IES and/or Complementarities

Paper	IES measurement	Factors influencing the returns from IES
Hitt et al. (2002)	Adoption levels of ERP system	
Hendricks et al. (2007)	Adoption of ERP/CRM/SCM	
Dong et al. (2009)	Adoption of SCM	Competitive intensity
Angst et al. (2011)	Health information technology	
Ho et al. (2011)	IT investment	Corporate governance
Aral et al. (2012)	Adoption of human capital management software	Human resource analytics, performance pay policy
Tambe et al. (2012)	IT usage based on IT employment	External focus, decentralization
Bardhan et al. (2013)	IT investment	R&D investment
Bloom et al. (2014)	ERP and data intranet	
Dranove et al. (2014)	Adoption of health IT	Local IT intensity
Tian and Xu (2015)	Scope of ERP system	Environmental uncertainty
Atasoy et al. (2016)	Adoption of ERP/CRM/Web applications	Firm size, average wage rate, industry technology intensity
Mithas and Rust (2016)	IT investment	IT strategy
Havakhor et al. (2019)	IT investment	R&D investment, advertising, environmental turbulence
Nagle (2019)	IT capital and open source software	IT intensity (ecosystem of complements)
Chau et al. (2020)	IT-business alignment	IT governance
Saldanha et al. (2020)	IT investment	IT-business alignment
Dhyne et al. (2021)	IT capital	Firm size
Huang et al. (2022)	IT knowledge spillovers	Absorptive capacity
This study	Adoption of ERP and CRM/SCM	ERP (CRM/SCM) module foundation, ERP-CRM/SCM sequential connection, continued ERP (CRM/SCM) adoption

Appendix B

We report the correlation among key variables used in this study in Appendix B. Because Panel A variables and Panel B variables in Table 2 are mainly defined at the observation level, we report the correlation table in Table B1 using the full sample. Because Panel C variables in Table 2 are firm specific and time invariant, we report the correlation table in Table B2 using one observation from each firm that went live with CRM/SCM.

Table B1. Correlation Table of Performance Variables and Adoption Variables

Variables	V1	V2	V3	V4	V5	V6	V7
Total assets (V1)	1						
Total equity (V2)	0.73	1					
Pretax income (V3)	0.41	0.50	1				
ERP purchase status (V4)	0.07	0.08	0.05	1			
ERP live status (V5)	0.09	0.13	0.07	0.56	1		
CRM/SCM purchase status (V6)	0.09	0.13	0.09	0.58	0.47	1	
CRM/SCM live status (V7)	0.09	0.16	0.12	0.31	0.56	0.54	1

Table B2. Correlation Table of IES Variables

Variables	V8	V9	V10	V11	V12	V13
Number of ERP modules (V8)	1					
No ERP-CRM/SCM going-live gap (V9)	-0.44	1				
Number of CRM/SCM modules (V10)	0.16	0.16	1			
Continued ERP adoption (V11)	-0.07	0.05	0.05	1		
Continued CRM/SCM adoption (V12)	0.09	0.01	-0.08	0.11	1	
Agile strategy level (V13)	-0.59	0.59	-0.27	0.40	0.48	1

We also report our classification of the industry in Table B3. We relied on the two-digit SIC code of firms to generate 10 industry categories.

Table B3. Industry Groups

Two-digit SIC	Industry	Description	Percentage in sample
1	1	Agriculture, forestry, fishing, and hunting	0.3
10.4-20	2	Mining and construction	2.1
20.11-29.11	3	Process manufacturing	22.6
30.11-39.9	4	High-tech manufacturing	35.6
40.13-50	5	Transportation, communications, electric, gas, and sanitary services	9.1
50.2-59.9	6	Wholesale trade and retail trade	6.8
60.2-67.99	7	Finance, insurance, and real estate	3.2
70.11-79.9	8	Personal services and business services	18.1
80.11-87.42	9	Legal, educational, and art services	1.6
99.97	10	Non-classifiable establishments	0.6

Appendix C

Appendix C focuses on the robustness checks on the average treatment effect. We investigated various concerns that could influence firm performance and CRM/SCM adoption. Table C1 summarizes the key concerns and our solutions.

Table C1. Alternative Explanations and Robustness Checks

No.	Concerns and/or alternative explanations	Solutions	Results
1	Standard errors might be clustered at the firm level rather than the industry level.	Use firm-level clustered standard errors	Table C2
2	Lagged effect of ERP (rather than CRM/SCM) increases firm performance.	Add lagged ERP dummies	Table C3
3	ROA and ROE are not the best measures of firm performance.	Use alternative firm performance measure Tobin's *q* and the Cobb-Douglas specification	Table C4
4	Firms that did not go live with CRM/SCM are not comparable to firms that went live with CRM/SCM.	Use coarsened exact matching to create a matched sample	Table C5
5	Many variables that can influence firm performance have not been controlled for.	Add five control variables into the model, e.g., number of employees, R&D expenses	Table C6
6	Going live with CRM/SCM is not random because it could be related to firm performance.	Use relative time model to check parallel trends assumption	Table C7
7	Going live with CRM/SCM is not random because it could be related to firm IT capability, or overall IT planning.	Use subsample firms that purchased CRM/SCM or firms that executed ERP	Table C8
8	Going live with of CRM/SCM is not random because it could be related to time-specific firm-wide improvement.	Control ERP implementation event and CRM/SCM implementation event	Table C9
9	ERP/CRM/SCM does not likely take effect in the first year.	Code CRM/SCM as 1 from the first full year after going live	Table C10
10	ERP/CRM/SCM does not likely take effect in the first year.	Use the next year's performance instead of current year's performance	Table C11

Table C2. Returns from ERP and CRM/SCM: Alternative Clustered Standard Errors

Dependent variable	ln(pretax income)	ln(pretax income)	ln(pretax income)	ln(pretax income)
Interpretation	ROA	ROA	ROE	ROE
CRM/SCM purchase status	-0.024 (0.060)		0.001 (0.060)	
CRM/SCM purchase action		-0.062 (0.053)		-0.057 (0.053)
CRM/SCM live status	0.108[a] (0.073)	0.104[b] (0.072)	0.111[c] (0.071)	0.113[d] (0.069)
Other control variables	√	√	√	√
Adjusted R^2	0.88	0.88	0.89	0.89
Observations	3,336	3,336	3,280	3,280

Note: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Standard errors are clustered at the firm level. *P*-values: a = 0.143, b = 0.147, c = 0.113, d = 0.103. Clustering at the firm level may unnecessarily amplify the standard errors given the nature of huge fluctuations in firm performance in our research context. Therefore, firm-level clustered standard errors should serve as a conservative estimation of the significance level. Because the economic magnitudes of impacts are around 10%, we still believe they are meaningful and sizeable, though not strictly significant.

Considering ERP is the prerequisite and foundation for CRM/SCM, the performance gains might actually come from ERP if ERP requires time to take real effect. To rule out this possibility, we applied two approaches. First, we included lead ERP variables to measure how the ERP go-live events one year and two years prior (i.e., L.ERP and L2.ERP) influenced firm performance in the current year. If ERP leads to a lagged effect, the coefficients of lead ERP variables should be positive and significant. Second, we generated a new continuous variable based on ERP go-live events, which measured the number of years since ERP went live. For example, if ERP went live in 2000, the variable took a value of 1 in 2000, 2 in 2001, 3 in 2002, and so on. If ERP takes effect slowly, this continuous variable should be able to capture that scenario. Table C3 reports the results, and the impacts of CRM/SCM are consistent.

Table C3. Returns from ERP and CRM/SCM: Controlling for the Lagged Effect of ERP				
Dependent variable	**ln(pretax income)**	**ln(pretax income)**	**ln(pretax income)**	**ln(pretax income)**
Interpretation	ROA	ROA	ROE	ROE
ERP live status	-0.029 (0.050)		-0.060 (0.045)	
L1. ERP live status	-0.062 (0.042)		-0.057 (0.045)	
L2. ERP live status	-0.002 (0.049)		-0.014 (0.049)	
Years since ERP went live		-0.011 (0.019)		-0.019 (0.014)
CRM/SCM purchase status	-0.025 (0.047)	-0.025 (0.047)	-0.000 (0.048)	-0.000 (0.048)
CRM/SCM live status	0.116*** (0.032)	0.090** (0.030)	0.119** (0.039)	0.085* (0.043)
Other control variables	√	√	√	√
Adjusted R^2	0.88	0.88	0.89	0.89
Observations	3,336	3,336	3,280	3,280

Note: ***p < 0.01; **p < 0.05; *p < 0.1. Standard errors are clustered at the industry level.

Table C4. Returns from ERP and CRM/SCM: Tobin's *q* Performance and Cobb-Douglas Specification		
Dependent variable	**Tobin's *q***	**ln(value added)**
Number of employees	-0.007* (0.004)	
ln(capital)		0.212*** (0.007)
ln(materials)		0.801*** (0.009)
ERP purchase status	-0.115 (0.114)	-0.040* (0.023)
ERP live status	0.598 (0.411)	-0.045** (0.020)
CRM/SCM purchase status	-0.814 (0.464)	0.032 (0.020)
CRM/SCM live status	0.265* (0.133)	0.050* (0.026)
Other control variables	√	√
Adjusted R^2	0.31	0.95
Observations	3,692	3,977

Note: ***p < 0.01; **p < 0.05; *p < 0.1. Standard errors are clustered at the industry level. See detailed definition and measurement of Tobin's *q* in Bardhan et al. (2013).

Table C5. Returns from ERP and CRM/SCM: CEM Matched Sample		
Dependent variable	**ln(pretax income)**	**ln(pretax income)**
Interpretation	ROA	ROE
CRM/SCM purchase status	0.141 (0.079)	0.177* (0.081)
CRM/SCM live status	0.188* (0.097)	0.196* (0.097)
Other control variables	√	√
Adjusted R^2	0.85	0.85
Observations	1,652	1,634

Note: ***p < 0.01; **p < 0.05; *p < 0.1. Standard errors are clustered at the industry level. We matched on ERP purchase year, CRM/SCM purchase year, sales (year 1997), and employees (year 1997). The assumption is these matched firms have similar performance and enterprise system plans before the go-live events of CRM/SCM.

Table C6. Returns from ERP and CRM/SCM: More Control Variables		
Dependent variable	**ln(pretax income)**	**ln(pretax income)**
Interpretation	ROA	ROE
CRM/SCM purchase status	-0.021 (0.049)	-0.001 (0.050)
CRM/SCM live status	0.095*** (0.028)	0.097** (0.039)
ln(cash)	0.055** (0.018)	0.050** (0.016)
ln(number of employees)	0.070 (0.043)	0.084* (0.038)
ln(selling and general expenses)	-0.026* (0.013)	-0.013 (0.018)
ln(advertising expenses)	0.020 (0.020)	0.037* (0.019)
ln(R&D expenses)	0.020 (0.034)	0.025 (0.034)
Other control variables	√	√
Adjusted R^2	0.88	0.89
Observations	3,336	3,280

Note: ***p < 0.01; **p < 0.05; *p < 0.1. Standard errors are clustered at the industry level. Because these additional control variables are not always available, we had to impute missing values with 0 and did not include them in the main analysis.

Table C7. Returns from ERP and CRM/SCM: Relative Time Model		
Dependent variable	**ln(pretax income)**	**ln(pretax income)**
Interpretation	ROA	ROE
CRM/SCM purchase status	-0.012 (0.042)	0.019 (0.047)
Relative time -5	0.106 (0.125)	0.134 (0.118)
Relative time -4	0.036 (0.074)	0.069 (0.079)
Relative time -3	0.042 (0.049)	0.068 (0.054)
Relative time -2	0.048 (0.054)	0.065 (0.049)
Relative time -1	baseline	baseline
Relative time 0	0.145** (0.064)	0.164*** (0.052)
Relative time 1	0.171** (0.060)	0.197** (0.062)
Relative time 2	0.039 (0.045)	0.036 (0.055)
Relative time 3	0.163 (0.132)	0.173 (0.136)
Relative time 4	0.180* (0.081)	0.175* (0.084)
Relative time 5	0.025 (0.367)	-0.015 (0.371)
Other control variables	√	√
Adjusted R^2	0.88	0.89
Observations	3,336	3,280

Note: ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$. Standard errors are clustered at the industry level. Relative time dummies reflect the year difference between the year of observation and the year of the go-live event. Lead dummies ranging from -5 to 2 indicate pre-treatment trends and lag dummies ranging from 0 to 5 indicate post-treatment impacts. Due to the length of the observation period, few observations are marked as "relative time 5," so the coefficient of "relative time 5" is inconclusive.

Table C8. Returns from ERP and CRM/SCM: Subsample Analysis		
Dependent variable	**ln(pretax income)**	**ln(pretax income)**
Interpretation	ROA	ROA
Sample	Firms that eventually purchased CRM/SCM	Firms that eventually went live with ERP
CRM/SCM purchase status	0.028 (0.057)	-0.004 (0.051)
CRM/SCM live status	0.081* (0.037)	0.089* (0.044)
Other control variables	√	√
Adjusted R^2	0.87	0.87
Observations	2,392	2,277

Note: ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$. Standard errors are clustered at the industry level. Column 1 uses a subsample including firms that eventually purchased CRM/SCM, and Column 2 uses a subsample including firms that eventually went live with ERP. The assumption is that these firms are more comparable because the only difference is whether they went live with CRM/SCM.

Table C9. Returns from ERP and CRM/SCM: Including Installment Events		
Dependent variable	**ln(pretax income)**	**ln(pretax income)**
Interpretation	ROA	ROE
CRM/SCM purchase status	-0.011 (0.048)	0.010 (0.050)
CRM/SCM installment status	-0.038 (0.045)	-0.029 (0.040)
CRM/SCM live status	0.108*** (0.030)	0.111** (0.036)
Other control variables	√	√
Adjusted R^2	0.88	0.89
Observations	3,336	3,280

Note: ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$. Standard errors are clustered at the industry level. Installment events are controlled in this analysis. If any events influence go-live events and firm performance simultaneously, those events would also likely influence installment events. Therefore, controlling for installment events could alleviate this specific simultaneity concern.

Table C10. Returns from ERP and CRM/SCM: Alternative Coding of Go-Live Events

Dependent variable	**ln(pretax income)**	**ln(pretax income)**
Interpretation	ROA	ROE
CRM/SCM purchase status	-0.011 (0.048)	0.013 (0.051)
CRM/SCM live status (alternative coding)	0.086** (0.036)	0.081** (0.032)
Other control variables	√	√
Adjusted R^2	0.88	0.89
Observations	3,336	3,280

Note: ***p < 0.01; **p < 0.05; *p < 0.1. Standard errors are clustered at the industry level. We used alternative coding of go-live events in this analysis. We coded live status as 1 starting from the first full year of go-live events rather than the first year of go-live events.

Table C11. Returns from ERP and CRM/SCM: Using Next Year's Performance

Dependent variable	**ln(pretax income)**	**ln(pretax income)**
Interpretation	ROA	ROE
CRM/SCM purchase status	-0.001 (0.044)	0.020 (0.048)
CRM/SCM live status	0.091** (0.037)	0.081** (0.033)
Other control variables	√	√
Adjusted R^2	0.88	0.89
Observations	3,336	3,280

Note: ***p < 0.01; **p < 0.05; *p < 0.1. Standard errors are clustered at the industry level. We use the next year's performance instead of the current year's performance in case IES adoption happens at the end of the focal year.

Appendix D

Appendix D focuses on the concerns that are specific to the IES implementation strategies and decisions. In the main text, we measured ERP module foundation with the number of modules and the number of module domains. However, we did not break down each domain in the main analysis because we didn't have enough number of firms in each category. Table D1 summarizes the detailed breakdown of ERP module foundation. The first column corresponds to the dummy categories used in Table 4 (i.e., ERP module domains—basic or two, ERP module domains—three, and ERP module domains—four). To estimate the impacts of detailed ERP module domains, we created several dummy variables according to Table D1 and marked the values of these dummy variables based on ERP module foundation. We then interacted these dummies with "CRM/SCM live status" and estimated their coefficients in Table D2. As a result of this analysis, the ultimate impact of each module foundation category comes from the sum of all coefficients of related interaction terms. For example, the ultimate impact of CRM/SCM on ROA with 3-a module foundation equals 0.250 + 0.172 + 0.446 + (-0.826), which is 0.042. We calculated the ultimate impacts of all module foundation categories and report them in the last two columns in Table D1. Our interpretation of the results is limited because only a few firms contributed to the estimation for several categories, but we can still conclude that the basic module foundation leads to higher returns from CRM/SCM compared to the all-inclusive (i.e., 4) module foundation.

Table D1. A Breakdown of ERP Module Domains

Module foundation	Count	Basic	HR	Finance (CO)	Operation (PP)	HR+ CO	HR+ PP	CO+P P	All	ROA impact	ROE impact
Basic	78	1	0	0	0	0	0	0	0	0.250***	0.265***
2-a	14	1	1	0	0	0	0	0	0	0.422	0.461
2-b	3	1	0	1	0	0	0	0	0	0.695***	0.255
2-c	1	1	0	0	1	0	0	0	0	-0.070	-0.152**
3-a	7	1	1	1	0	1	0	0	0	0.042	-0.077
3-b	1	1	1	0	1	0	1	0	0	1.219***	1.206***
3-c	41	1	0	1	1	0	0	1	0	-0.039	0.001
4	66	1	1	1	1	1	1	1	1	-0.025	-0.020

Table D2. Returns from ERP and CRM/SCM: ERP Module Domains

Dependent variable	**ln(pretax income)**	**ln(pretax income)**
Interpretation	ROA	ROE
CRM/SCM purchase status	-0.029 (0.048)	-0.003 (0.047)
CRM/SCM live status-basic	0.250*** (0.055)	0.265*** (0.046)
CRM/SCM live status—with HR domain	0.172 (0.325)	0.196 (0.347)
CRM/SCM live status—with CO domain	0.446*** (0.073)	-0.010 (0.451)
CRM/SCM live status—with PP domain	-0.319*** (0.078)	-0.417*** (0.063)
CRM/SCM live status—with both HR and CO domains	-0.826** (0.292)	-0.528 (0.548)
CRM/SCM live status—with both HR and PP domains	1.117*** (0.338)	1.162*** (0.351)
CRM/SCM live status—with both PP and CO domains	-0.415*** (0.113)	0.163 (0.464)
CRM/SCM live status—with HR, CO, and PP domains	-0.449 (0.363)	-0.851 (0.579)
Other control variables	√	√
Adjusted R^2	0.88	0.89
Observations	3,336	3,280

Note: ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$. Standard errors are clustered at the industry level.

When we investigated the impacts of IES implementation strategies on the returns from IES, we first developed a concept called "agile strategy level," which was based on the five IES implementation decisions. We summarize the definition of each agile decision and the number of firms that went with each decision in Table D3. Based on the concept of "agile strategy level," we further developed three categories of IES implementation strategies, called the phased strategy, the semi-agile strategy, and the agile strategy. Table D4 summarizes the mapping between various agile strategy levels and the operationalized IES implementation strategies.

Table D3. IES Implementation Decisions and Agile Strategy Levels

Decision	ERP module foundation	ERP-CRM/SCM sequential connection	CRM/SCM module foundation	Continued ERP adoption	Continued CRM/SCM adoption
Agile strategy definition	Basic or two ERP module domains	Less than one-year gap between ERP and CRM/SCM go-live events	Only one CRM/SCM module domain	Adopted new ERP modules after going live with CRM/SCM	Adopted new CRM/SCM modules after going live with CRM/SCM
Firms identified as agile	96	126	178	29	61
Firms identified as not agile	115	85	33	182	150

Table D4. Agile Strategy Levels and IES Implementation Strategies

Agile strategy level	0	1	2	3	4	5
Count of firms	3	55	59	65	23	6
IES implementation strategy	Phased		Semi-agile		Agile	
Number of firms	58		124		29	